MILITARY SPECIFICATION SHEET

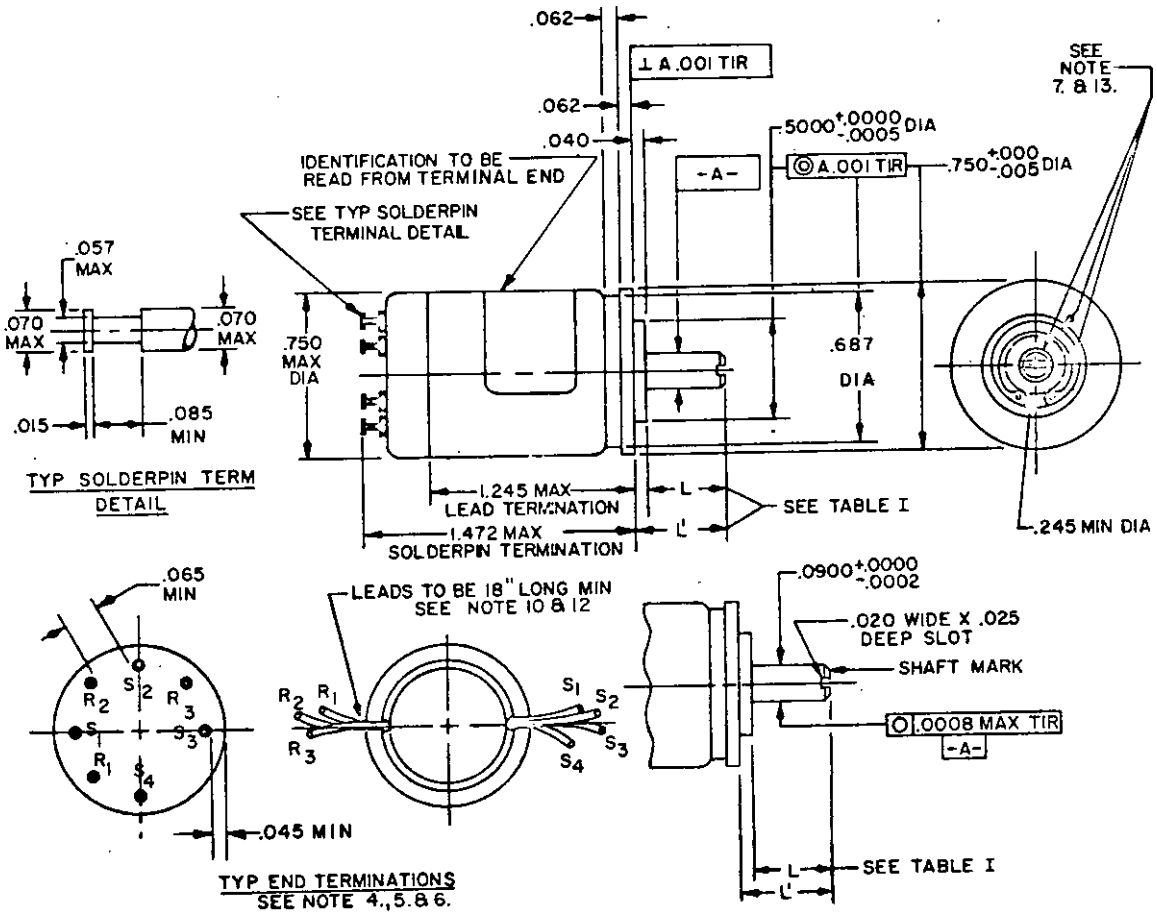
RESOLVER, ELECTRICAL

TYPE 08R7N4A

These resolvers are for replacement purposes only, not recommended for new designs.

The complete requirements for procuring the resolver described herein shall consist of this document and the latest issue of MIL-R-23417.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

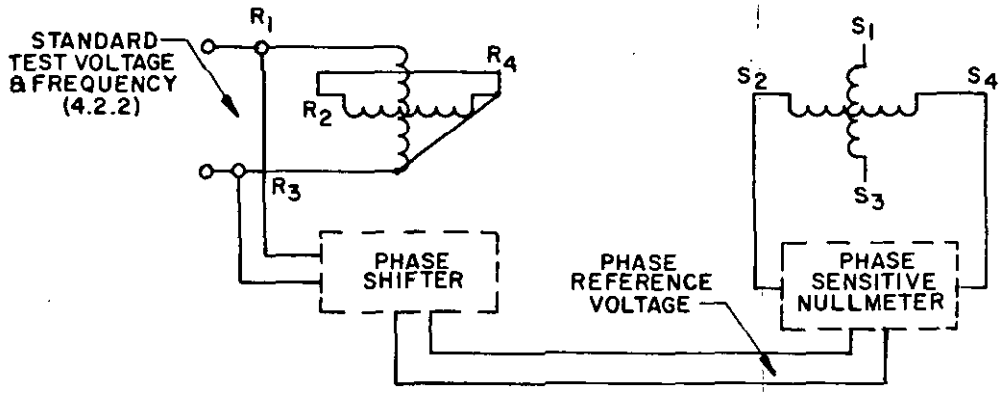


FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. Terminal lead wires shall depart through the back surface of the terminal end of the resolver in any acceptable manner.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Solder pin terminal stress shall be tested in accordance with MIL-STD-202, Method 211, as a qualification test. The terminals shall be capable of withstanding, without damage or loosening, a steady pull of the force specified in any direction by a wire fastened in the normal manner. Internal connections to terminals shall be welded or soldered with high temperature solder SN-5 per QQ-S-571.
7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. resolver zero, and will bear no relation to the center lines shown on the drawing.
8. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.
9. For information only see synchro mounting clamp MS17183 or equal.
10. Leads shall be type E-28.
11. A voltage value of .5 to 26V shall be entered in the 'Volts' designation space on the identification plate.
12. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.
13. The front retainer shall be threaded or a snap ring.
14. The following items are exceptions to MIL-R-23417.
    a. Resolver zero - Use diagram below in lieu of figure 1 in MIL-R-23417.



b. Electrical angle - Use information below in lieu of 3.4.2.2 in MIL-R-23417.



$E(S1-S3) = KE(R1-R3)\cos\theta - KE(R2-R4)^*\sin\theta$

$E(S2-S4) = KE(R2-R4)^*\cos\theta + KE(R1-R3)\sin\theta$

Where K = Transformation ratio.

*Designation R4 to be used for analytic purposes only. For testing purposes use designation R3 here and throughout MIL-R-23417 where applicable.

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RESOLVER, ELECTRICAL: TYPE 08R7N4A

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirements	Value	Unit	Tolerance
1. Excitation winding	Rotor	---	---
2. Rated frequency	400	Hz	±1%
3. Input voltage:			
Maximum	26	Volts rms	±1%
Test	26	Volts rms	±1%
4. Primary current	43	Milliamps	Nominal
5. Primary power	.45	Watts	Nominal
6. DC resistance:			
Rotor	230	Ohms	Reference
Stator	170	Ohms	Reference
7. Impedance:			
Zss	448 + j175	Ohms	Reference
Zro	270 + j630	Ohms	R±15%, +jx±10%
Zso	250 + j830	Ohms	R±15%, +jx±10%
8. Transformation ratio Output/input	1.000		2%
9. Equality of transformation ratio Output/input	±1	Percent	Maximum
10. Phase shift Output/input	20	Degrees	±2
11. Accuracy Electrical error	±5	Minutes	Maximum
12. Null voltage:			
Fundamental	20.8	Mv rms	Maximum
Total	28.6	Mv rms	Maximum
13. Friction torque:			
@23°C	.04	Oz-In	Maximum
@ -55°C and +125°C	0.1	Oz-In	Maximum
14. Radial play	.0004 (For reversal of 1/2 lb load)	Inches	Maximum
15. End play	.0001-.0015 (For reversal of 1/2 lb load)	Inches	Min-Max
16. Wire lead stress (see note 12)	2	Pound	Minimum
17. Terminal stress (see note 6)	3	Pound	Minimum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table I.

Example:

M23417/ 38 -01

Basic document number — Spec sheet number — Dash number from table I

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TABLE I. Dash numbers and characteristics.

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				L (ref)	L' - dim to first mounting flange surface (±.010)
-01	.090	Plain	Wire leads	.325	.365
-02	.090	Plain	Wire leads	.406	.446
-03	.090	Plain	Wire leads	.500	.540
-04	.090	Plain	Solder pins	.325	.365
-05	.090	Plain	Solder pins	.406	.446
-06	.090	Plain	Solder pins	.500	.540

Custodians:
Army - MU
Navy - AS
Air Force - 80

Review activities:
Army - MU, EL
Navy - AS, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - MI, ME, AV
Navy - CG, SH
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

(Project 5990-0254)

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